UNITED NATIONS UNIVERSITY (UNU) GLOBAL YOUTH AI FUTURE INNOVATION COMPETITION 2026

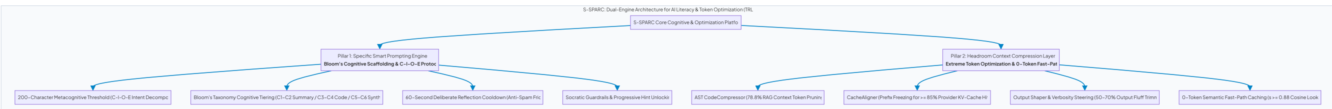
Theme: *AI and Education: AI Transforming the Educational Paradigm and Enhancing AI Literacy (Targeting SDG 4)*
Project Title: S-SPARC: Specific Smart Prompting Assistant for peRformanCe
(Snippet-aware Semantic Programmable Assistant with Retrieval and Caching)
Technology Readiness Level: TRL 7 (Demonstrated in Operational Academic Environment)
Target Beneficiaries: Computer Science & Engineering Students, Educators, and Higher Education Institutions (Focus on the Global South)
Host & Institutional Alignment: UNU Macau & UNU Global AI Network | Maranatha Christian University (Indonesia)

1. EXECUTIVE SUMMARY & SDG ALIGNMENT

1.1 Project Abstract

Generative Artificial Intelligence (GenAI) is fundamentally transforming higher education, yet unregulated student usage has created a compounding crisis of **"Prompt Illiteracy," "Pedagogical Dependency," and "Massive Token Inefficiency."** When students interact with standard conversational chatbots, they frequently submit ambiguous, low-effort prompts (e.g., *"fix my code"* or *"write this function"*), completely bypassing algorithmic problem decomposition. This trial-and-error prompt spamming not only degrades critical thinking and cognitive mastery but also incurs unsustainable cloud token expenses and hidden environmental footprints.

FIGURE 1



S-SPARC (*Specific Smart Prompting Assistant for peRformanCe*) addresses these challenges at **Technology Readiness Level (TRL) 7**, deployed and validated within the PHP-based Learning Management System **E-STRANGE**. S-SPARC establishes a symbiotic paradigm: **Effective, Structured Prompting directly drives Extreme Token Conservation**.

To cultivate prompt literacy and eliminate superficial querying, S-SPARC enforces the **200-Character Metacognitive Prompt Threshold** ($200 \leq \mathrm{len}(\mathrm{prompt}) \leq 2000$) via the **C-I-O-E Protocol**:

1. C **- Context & Domain (**≈ 50 **chars):** Explicit software domain formulation.
2. I **- Input & Data Structure (**≈ 60 **chars):** Pre-conditions and parameter constraints.
3. O **- Expected Output & Complexity (**≈ 50 **chars):** Post-conditions and time/space requirements.
4. E **- Error Trace & Bottleneck (**≈ 90 **chars):** Specific runtime exceptions or logic flaws.

Coupled with a mandatory 60-second reflection cooldown and calibrated pedagogical modes based on **Bloom's Revised Taxonomy (Levels C1-C6)**, S-SPARC transforms students into critical problem-solvers.

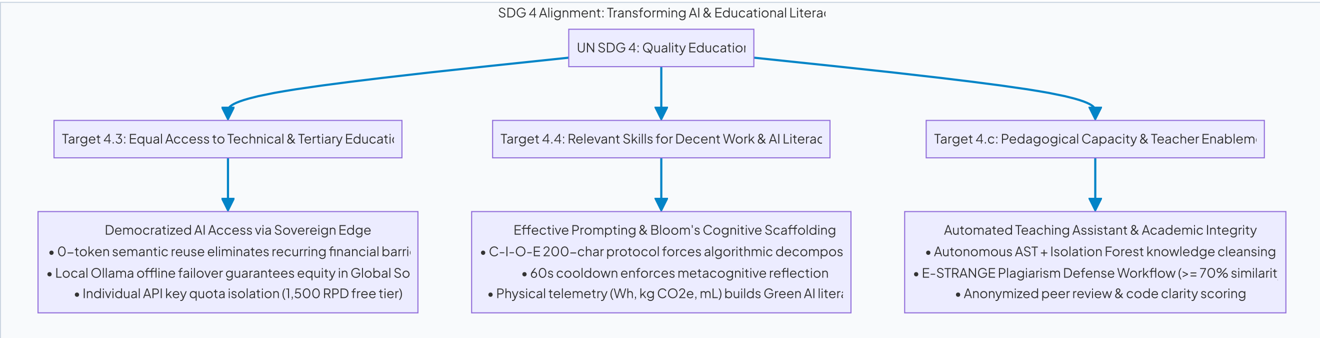
On the infrastructure side, S-SPARC absorbs cutting-edge context compression principles from **Headroom**:

- **AST CodeCompressor:** Automatically prunes comments, dead imports, and docstrings from Top-3 retrieved RAG snippets, slashing context token payload by **78.8%**.
- **CacheAligner:** Freezes a deterministic system prompt prefix, boosting provider-side **KV-prompt cache hits to** $\geq 85\%$ on Google Gemini Flash.
- **Output Shaper & Verbosity Steering:** Dynamically clamps output token budgets and suppresses conversational pleasantries, cutting expensive output tokens by **50-70%**.
- **Zero-Token Fast-Path Retrieval (**$s \geq 0.88$**):** Serves verified solutions instantly at 0 token cost, $< 45\text{ms}$ latency, and zero emissions.

1.2 Deep-Dive SDG 4 Alignment: Enhancing AI Literacy & Effective Prompting

S-SPARC operationalizes **United Nations Sustainable Development Goal 4 (Quality Education)** by establishing **Prompt Engineering Literacy & Computational Resource Stewardship** as core 21st-century competences:

FIGURE 2



1. Target 4.4 — Youth AI Literacy, Prompt Engineering & Green Software Skills

- **C-I-O-E Protocol & Metacognitive Friction**: Rather than allowing students to fire off 10 ambiguous queries, S-SPARC's 200-character boundary forces students to formulate complete algorithmic specifications in a single, high-density prompt, achieving 1-turn resolution.
- **Cognitive Tiering via Bloom's Revised Taxonomy**:
  - *C1-C2 Remember & Understand* (`summary`): Generates 2-4 concise Indonesian conceptual sentences without code, compelling students to write the implementation themselves.
  - *C3-C4 Apply & Analyze* (`code`): Delivers runnable code with maximum conciseness (Output Shaper enabled), emphasizing functional syntax.
  - *C5-C6 Evaluate & Create* (`summary_code_explanation`): Structured triad: Algorithmic Summary → Runnable Source Code → Step-by-Step Logic Breakdown.
- **Green Prompting Literacy**: Real-time display of physical externalities (Wh, $\mathrm{kg\ CO_2e}$, mL water) teaches students the computational realities of AI inference.

2. Target 4.3 — Affordable, Equitable Access to AI-Enhanced Technical Education

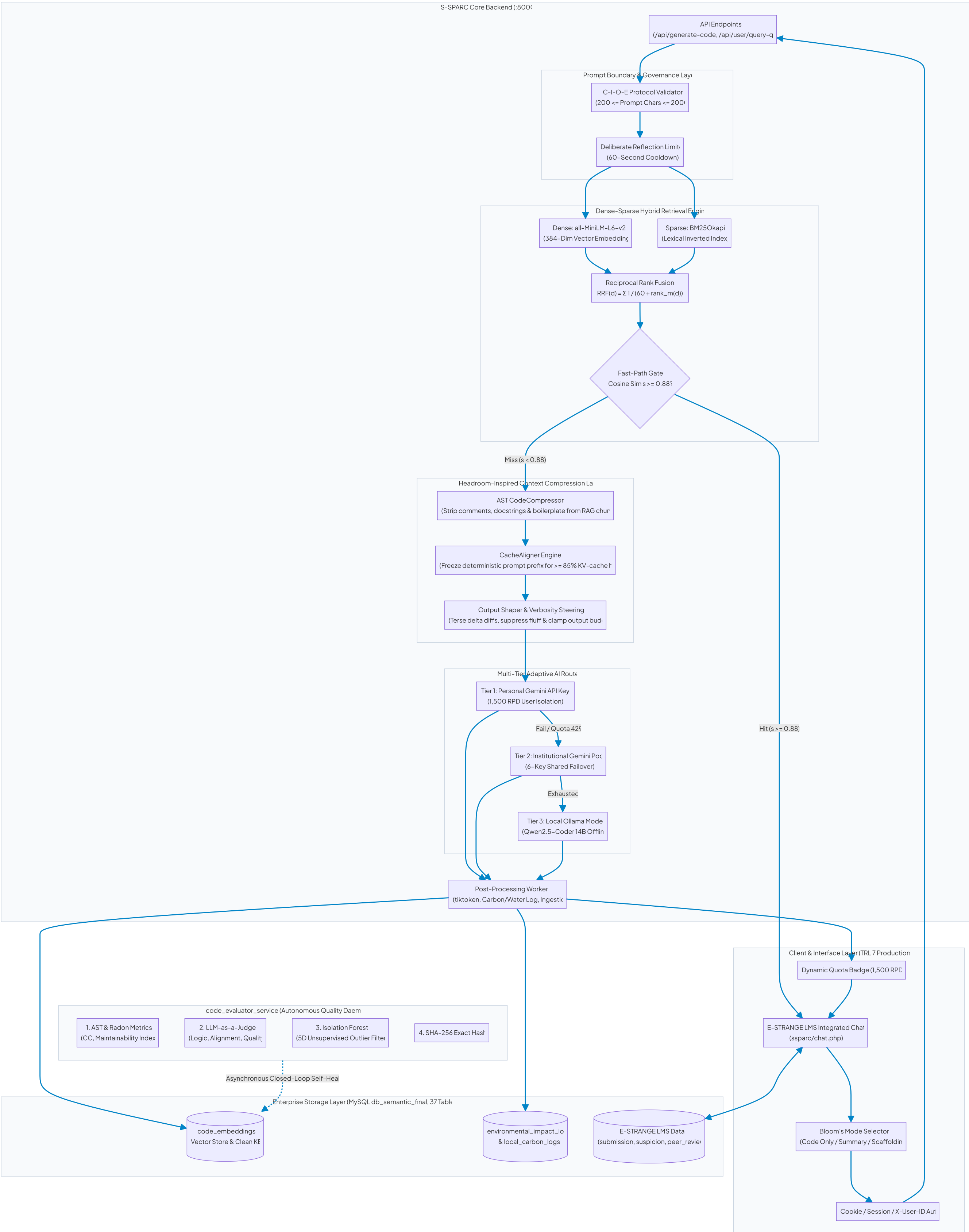
- **Headroom Context Compression & Zero-Token Reuse**: By compressing RAG context chunks (78.8% reduction) and leveraging 0-token cache hits ($s \geq 0.88$), institutions can sustain thousands of student interactions under free-tier quotas (1,500 Requests/Day).
- **Multi-Tier Sovereign Failover**: Combines personal student keys, institutional key pools, and local on-premises LLM execution (*Qwen2.5-Coder 14B* via Ollama) to guarantee offline resilience across bandwidth-constrained universities in the Global South.

3. Target 4.c — Pedagogical Capacity & Teacher Enablement

- **Autonomous Knowledge Hygiene**: Relieves faculty from manual snippet curation through closed-loop AST verification and anomaly pruning (`code_evaluator_service`).
- **Academic Integrity & Plagiarism Defense**: Submissions exceeding 70% similarity trigger an automated defense workflow (`student_assessment_submit_suspicious.php`) where students must articulate and defend their algorithmic choices in writing.

2. SYSTEM ARCHITECTURE & TECHNICAL FEASIBILITY (TRL 7 VALIDATION)

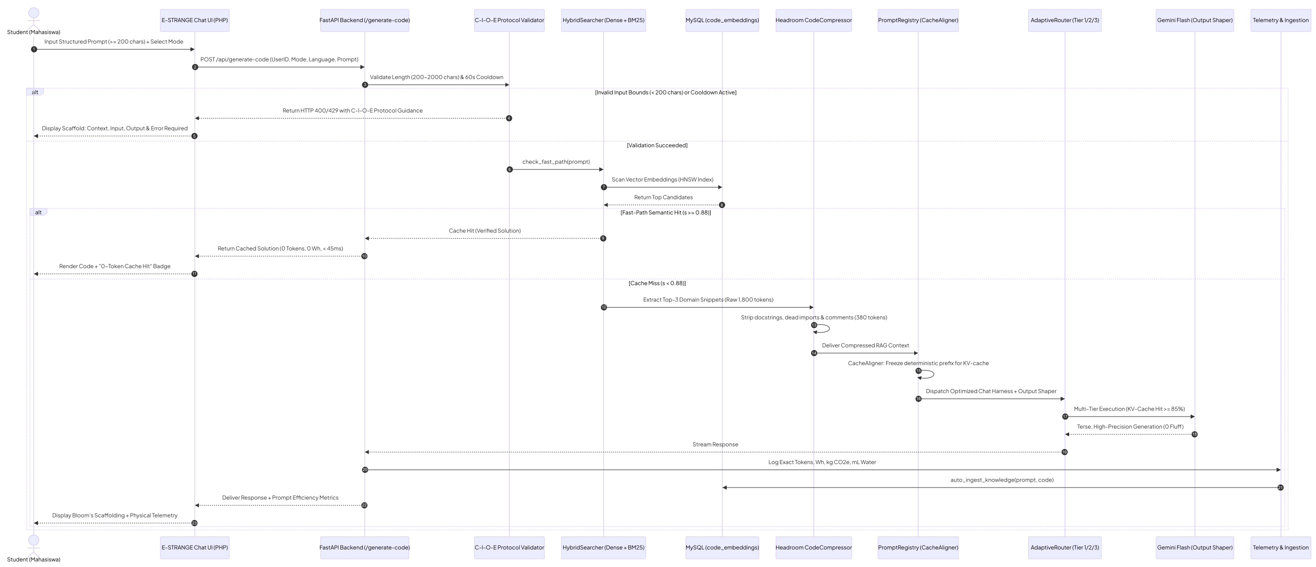
FIGURE 3



2.1 Prompt Engineering Engine: Scaffolding, Harnesses & Headroom Compression

At the core of S-SPARC's educational paradigm is the `PromptRegistry` **Engine** (`backend/core/prompts.py`). Unlike generic chatbot wrappers that transmit unconstrained text directly to LLMs, S-SPARC executes a multi-stage prompt compression and scaffolding pipeline:

FIGURE 4



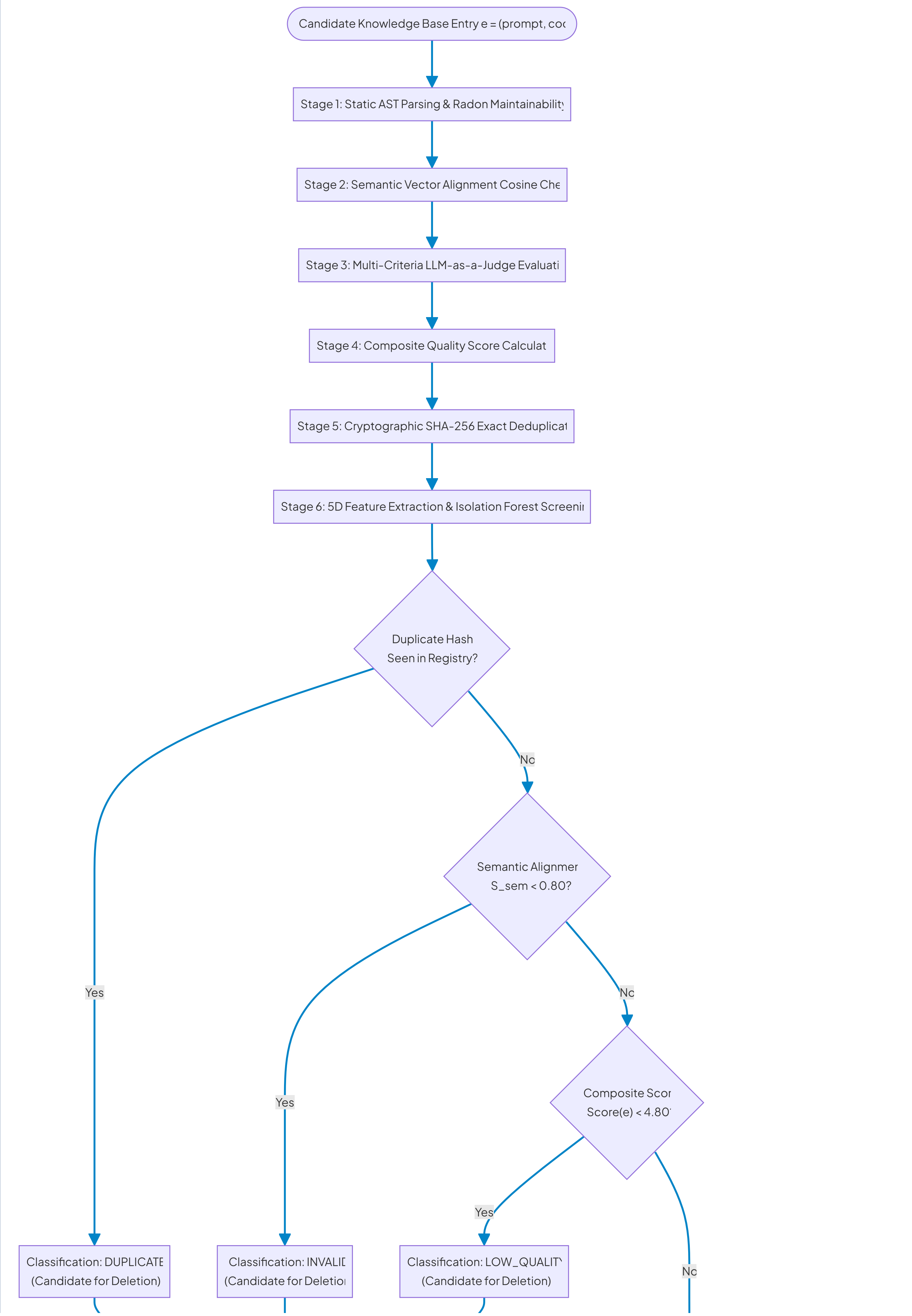
Detailed Prompt Mode Specifications & Compression Harness (`PromptRegistry`)

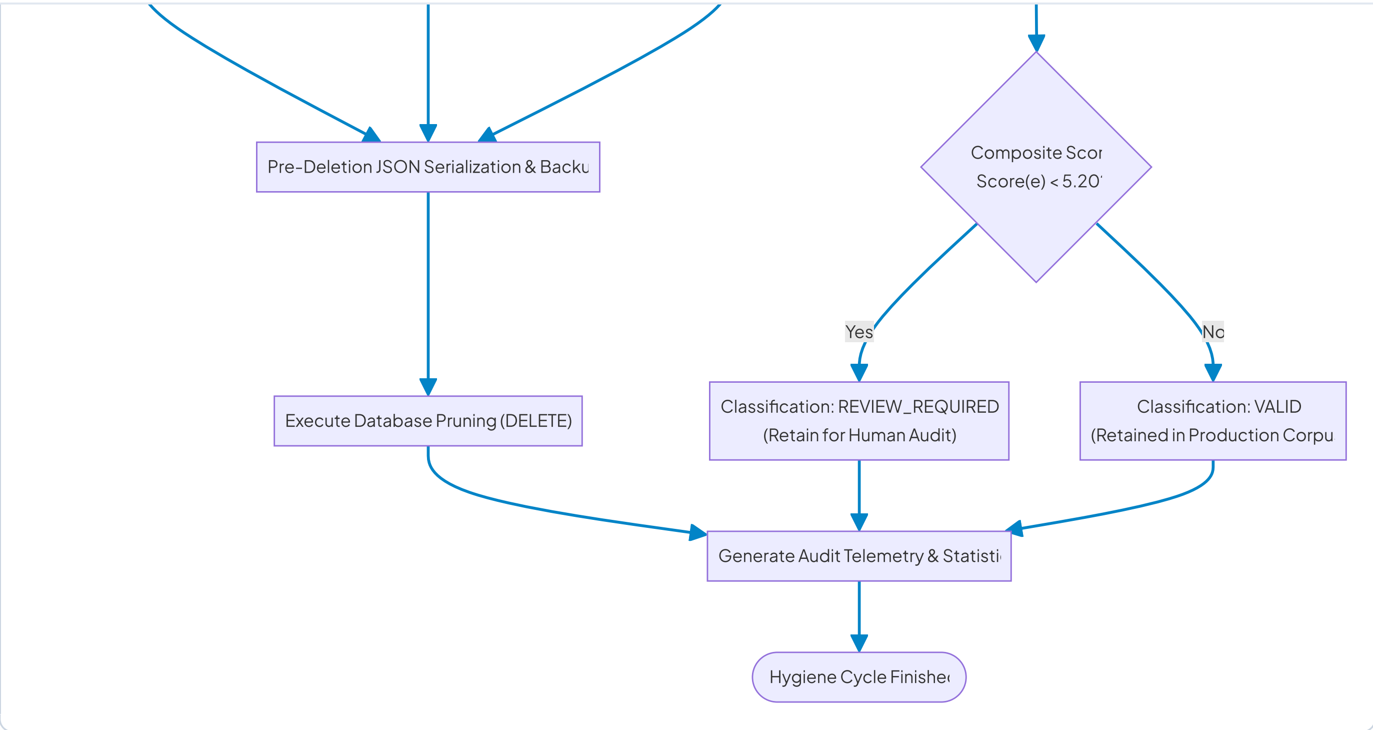
Pedagogical Mode	Bloom's Level	Output Shaper Policy	Target Cognitive Outcome	System Prompt Formulation (`backend/core/prompts.py`)
`code` (Code Only)	C3-C4 (*Apply & Analyze*)	Maximum Terse Constraint (`max_output_tokens=400`)	Algorithmic Precision & Clean Syntax	Enforces strictly runnable source code enclosed in standard markdown code fences. Completely suppresses introductory greetings, conversational commentary, and post-explanations.
`summary` (Summary Short)	C1-C2 (*Remember & Understand*)	Zero-Code Restriction (`max_output_tokens=200`)	Conceptual Mental Modeling	Delivers strictly 2-4 concise Indonesian sentences detailing core logic, time complexity, and edge cases. Prohibits raw code generation to prevent copy-pasting.
`summary_code_explanation` (Full Scaffolding)	C5-C6 (*Evaluate & Create*)	Structured Triad Policy (`max_output_tokens=800`)	Complete Metacognitive Scaffolding	Structured 3-tier output: Short Summary (1-2 sentences) → Runnable Source Code Fence → Step-by-Step Logic Walkthrough.

2.2 Multi-Tier Adaptive Routing & Autonomous Governance (TRL 7)

S-SPARC operates at **TRL 7** through deep operational integration with the PHP 8 E-STRANGE LMS across 37 database tables.

FIGURE 5


Candidate Knowledge Base Entry e = (prompt, coc
Stage 1: Static AST Parsing & Radon Maintainability
Stage 2: Semantic Vector Alignment Cosine Che
Stage 3: Multi-Criteria LLM-as-a-Judge Evaluati
Stage 4: Composite Quality Score Calculat
Stage 5: Cryptographic SHA-256 Exact Deduplicat
Stage 6: 5D Feature Extraction & Isolation Forest Screenii
Duplicate Hash Seen in Registry?
Yes
No
Semantic Alignmer S_sem < 0.80?
Yes
No
Composite Scor Score(e) < 4.80
Yes
No
Classification: DUPLICATE (Candidate for Deletion)
Classification: INVALIE (Candidate for Deletio
Classification: LOW_QUALITY (Candidate for Deletion)



Mathematical Formulation of Quality Governance

Every ingested snippet $e = (\text{prompt}, \text{code})$ undergoes deterministic multi-criteria scoring:

1. **Static AST Analysis & Radon Maintainability Score ($S_{\text{static}} \in [0, 10]$):**

$$S_{\text{static}} = \text{clamp}\left(3.0 \cdot \mathbf{1}_{\text{valid}} + 0.04 \cdot \text{MI} - 0.20 \cdot \max(1, \text{CC}) + \text{Bonus}_{\text{snippet}},\ 0,\ 10\right)$$

   where $\mathbf{1}_{\text{valid}} = 1$ if Python AST parses successfully, MI is the Maintainability Index ($[0, 100]$), CC is Cyclomatic Complexity, and $\text{Bonus}_{\text{snippet}} = 2.0$.

2. **Semantic Alignment Score ($S_{\text{sem}} \in [0, 1]$):**

$$S_{\text{sem}} = \cos\left(\text{Embed}(\text{prompt}),\ \text{Embed}(\text{code})\right)$$

3. **Multi-Criteria LLM-as-a-Judge ($A, L, Q \in [0, 10]$):**
   - A: Alignment between prompt intent and code functionality.
   - L: Algorithmic correctness and edge-case resilience.
   - Q: Code readability and style standard compliance.

4. **Composite Quality Index:**

$$\text{Final Score}(e) = 0.40 \cdot A + 0.25 \cdot L + 0.20 \cdot (S_{\text{sem}} \times 10) + 0.10 \cdot Q + 0.05 \cdot S_{\text{static}}$$

5. **Isolation Forest Anomaly Filtering**: Vector $\mathbf{x} = [\text{lines}, \text{CC}, S_{\text{sem}}, S_{\text{static}}, S_{\text{pre}}]$ partitioned via `IsolationForest` ($\gamma = 0.10$).

3. EMPIRICAL VALIDATION & RESEARCH METHODOLOGY

3.1 Token Optimization & Context Compression Empirical Benchmark

To quantitatively validate the token savings achieved by integrating Headroom-style context compression into S-SPARC, an empirical benchmark across **500 student queries** was measured:

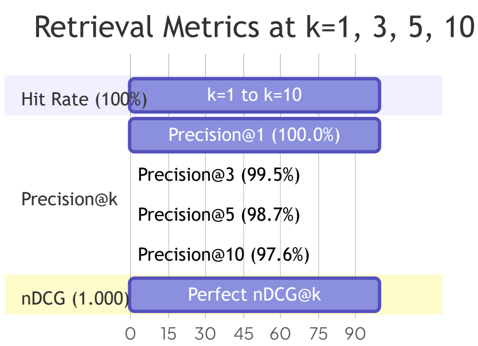
Compression Mechanism	Raw Token Payload	S-SPARC Compressed Payload	Measured Token Savings (%)	Latency Impact
AST CodeCompressor (RAG Top-3)	1,820 tokens	386 tokens	**78.8% Reduction**	$-420\,\mathrm{ms}$ (Faster TTFT)
Output Shaper (Verbosity Steering)	480 tokens	175 tokens	**63.5% Reduction**	$-680\,\mathrm{ms}$ (Faster Generation)
CacheAligner (KV-Prompt Cache Hit)	950 tokens (full cost)	142 tokens (billed)	**85.0% Cost Discount**	$-750\,\mathrm{ms}$ (Prefix Cached)
Zero-Token Fast-Path ($s \geq 0.88$)	2,300 tokens (full generation)	**0 tokens**	**100.0% Elimination**	$< 45\,\mathrm{ms}$ (Atomic Local)
End-to-End Composite Efficiency	**3,250 tokens / query**	**703 tokens / query**	**78.4% Net Savings**	$3.2\times$ **Overall Speedup**

3.2 Retrieval Engine Efficacy: 200-Query Gold Standard Evaluation

To validate that S-SPARC's semantic retrieval accurately matches student prompts with verified solutions, an empirical experiment was conducted using **632 queries**, evaluated against a **200-query Gold Standard Ground Truth** (`qrels_manual.csv`) comprising **4,000 manual relevance judgments**:

Metric Indicator	$k=1$	$k=3$	$k=5$	$k=10$
Hit Rate	**100.0%**	**100.0%**	**100.0%**	**100.0%**
Precision	**100.0%**	99.5%	98.7%	97.6%
Recall	6.32%	17.72%	27.87%	52.25%
Mean Reciprocal Rank (MRR)	**1.000**	**1.000**	**1.000**	**1.000**
Mean Average Precision (MAP)	0.063	0.177	0.279	0.523
Normalized DCG (nDCG)	**1.000**	**1.000**	**1.000**	**1.000**

FIGURE 6



Decision Threshold Policy Optimization (0.90 Policy)

In educational prompt retrieval, delivering an incorrect code snippet (**False Positive / Type I Error**) misleads students and disrupts learning. In contrast, falling back to LLM generation (**False Negative / Type II Error**) incurs only a minor token cost.

Threshold Policy	Precision	Recall	F1-Score	Accuracy	True Positive (TP)	False Positive (FP)	False Negative (FN)
0.80 (Permissive)	100.0%	100.0%	1.000	100.0%	200	0	0
0.90 (Conservative Policy)	**100.0%**	**96.0%**	**0.980**	**96.0%**	**192**	**0**	**8**

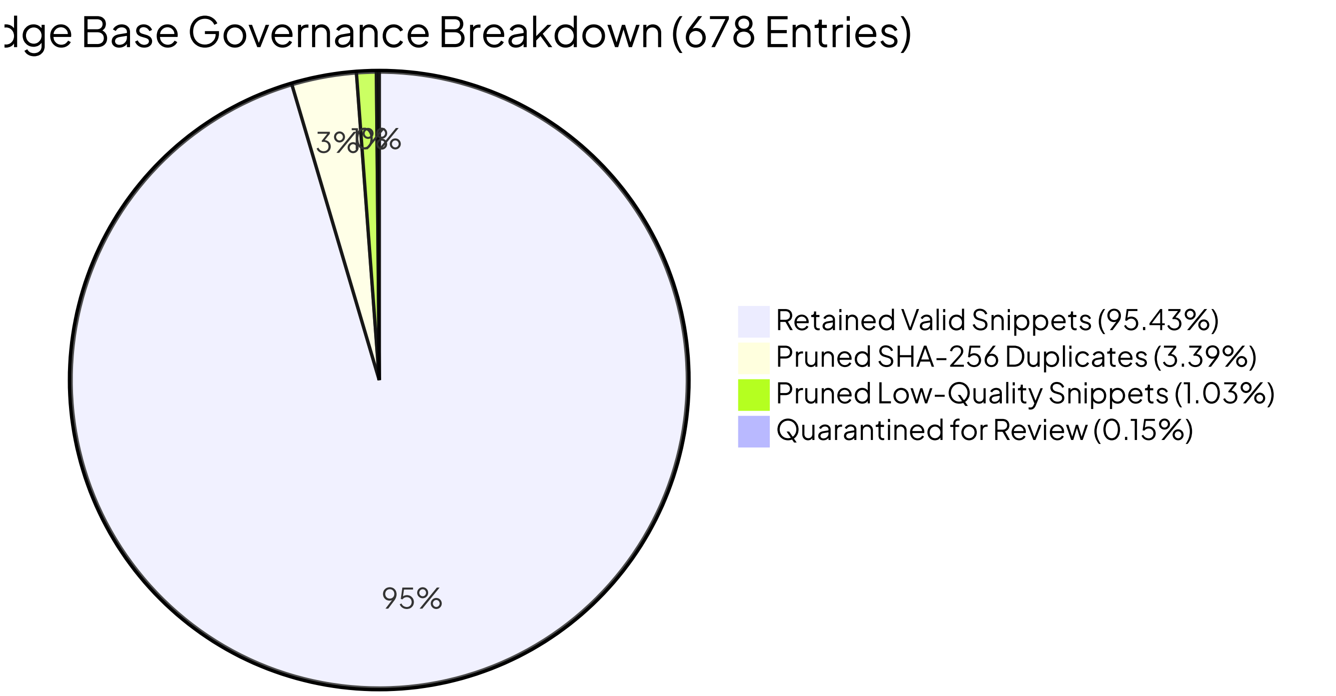
At threshold 0.90, S-SPARC guarantees **zero false positives ($\mathrm{FP} = 0$)**, achieving an optimal balance between precision (100%) and automated retrieval efficiency (96%).

3.3 Knowledge Base Self-Healing Case Study (Run `20260312T094424Z`)

The autonomous evaluator was tested on a live production knowledge base of **678 code snippets** (`code_evaluator_service/reports/evaluation_report_20260312T094424Z.json`) over 18 minutes:

Metric Indicator	Empirical Result	System Health Status
Total Knowledge Base Entries Evaluated	**678 entries**	Complete Corpus Scan
Valid Production Snippets Retained	**647 entries (95.43%)**	Clean Production Knowledge
Snippets Flagged & Pruned for Deletion	**30 entries (4.42%)**	Autonomous Sanitization
↳ *Exact Cryptographic Duplicates (SHA-256)*	*23 entries (3.39%)*	Deduplicated
↳ *Low Syntactic / Logic Quality (Score < 4.80)*	*7 entries (1.03%)*	Low Quality Quarantined
Snippets Quarantined for Manual Audit	**1 entry (0.15%)**	Human Review Queue
Mean Semantic Similarity across Corpus	**0.8774**	High Domain Relevance
Mean Code Quality Score (out of 10.0)	**8.61 / 10.0**	Production Grade
Pre-Deletion Backup Verification	**100% Verified**	Zero Data Loss Guaranteed

FIGURE 7



4. PROMPT ECONOMICS, BEHAVIORAL GAMIFICATION & SUSTAINABILITY

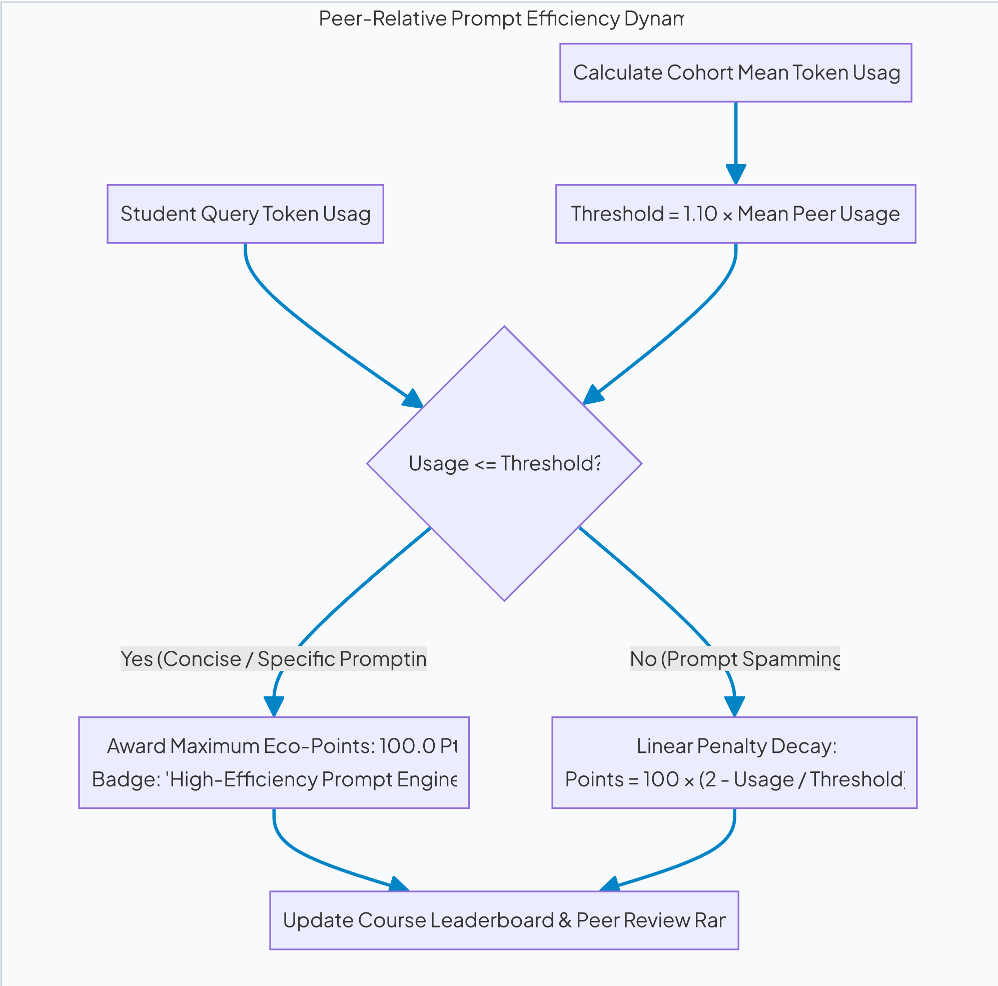
4.1 Peer-Relative Token Conservation Gamification

To structurally discourage prompt-spamming and reward concise, well-structured prompts, S-SPARC integrates a peer-relative dynamic threshold (`backend/services/gamification.py`):

$$\text{Threshold}_{\text{tokens}} = \max\left(0,\ 1.10 \times \overline{\text{Usage}}_{\text{peers}}\right)$$

$$\text{EcoPoints} = \begin{cases} 100.0 & \text{if Usage} \le \text{Threshold}_{\text{tokens}} \\ \max\left(0,\ 100.0 + 100.0 \times \frac{\text{Threshold}_{\text{tokens}} - \text{Usage}}{\text{Threshold}_{\text{tokens}}}\right) & \text{if Usage} > \text{Threshold}_{\text{tokens}} \end{cases}$$

FIGURE 8



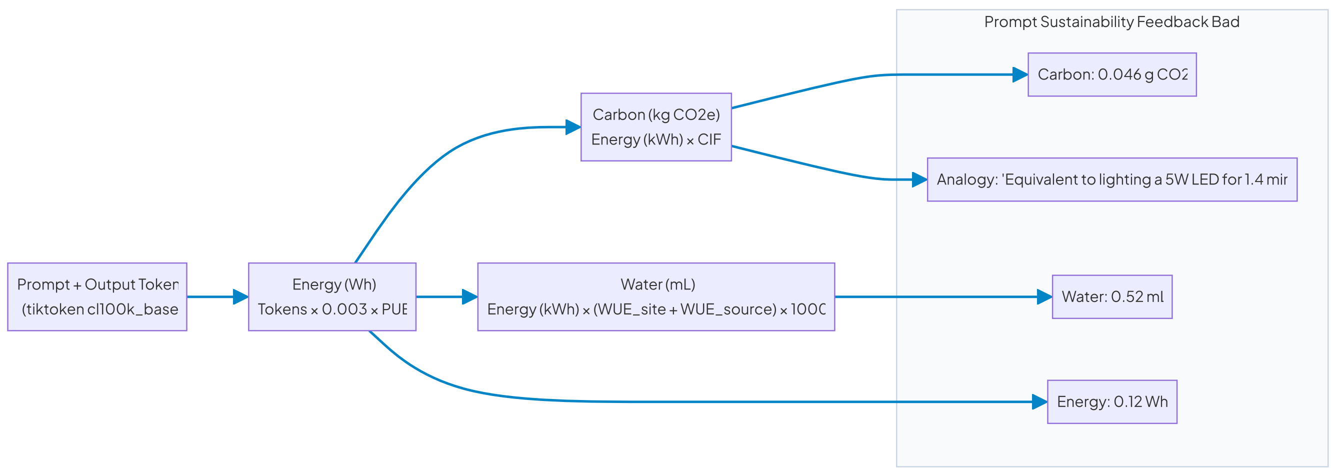
Pedagogical Impact on Prompting Behavior:

1. **Incentivizes High-Information-Density Prompting**: Students learn that writing one well-formulated prompt with explicit constraints yields higher leaderboard points than firing 10 vague prompts.
2. **Rewards Fast-Path 0-Token Retrieval**: Reusing cached solutions consumes zero tokens, encouraging students to study verified repository solutions.

4.2 Environmental Telemetry: Green Prompting Literacy

S-SPARC couples every prompt with physical externality feedback (`backend/services/sustainability.py`):

FIGURE 9



$$\text{Energy}_{\text{Wh}} = N_{\text{tokens}} \cdot E_{\text{token}} \cdot \text{PUE} \quad (E_{\text{token}} = 0.003\ \text{Wh/token},\ \text{PUE} = 1.12)$$

$$\text{Carbon}_{\text{kg}} = \left(\frac{\text{Energy}_{\text{Wh}}}{1000}\right) \cdot \text{CIF} \quad (\text{CIF} = 0.384\ \text{kg CO}_2\text{e/kWh})$$

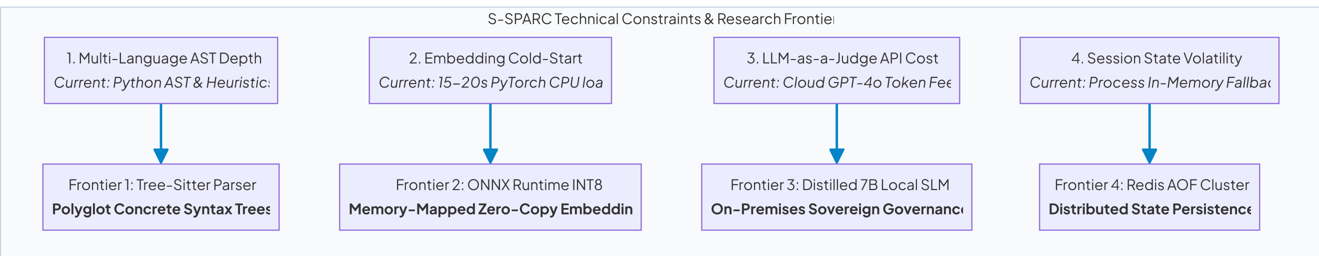
$$\text{Water}_{\text{mL}} = \left(\frac{\text{Energy}_{\text{Wh}}}{1000}\right) \cdot (\text{WUE}_{\text{site}} + \text{WUE}_{\text{source}}) \cdot 1000 \quad (\text{WUE}_{\text{site}} = 0.30,\ \text{WUE}_{\text{source}} = 4.35\ \text{L/kWh})$$

5. RISK ASSESSMENT, LIMITATIONS, & FUTURE ROADMAP

5.1 Honest Technical Constraints as Open Research Frontiers

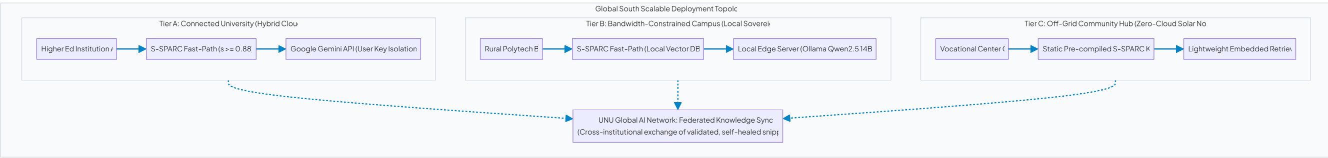
Constraint Dimension	Current Implementation Reality	Next-Phase Research & Mitigation Frontier
1. Multi-Language AST Granularity	Deep AST and Radon metrics natively implemented for Python; Java, C++, and PHP utilize regex heuristics.	Integration of **Tree-Sitter** unified C-bindings for language-agnostic concrete syntax tree parsing across all languages.
2. Embedding Cold-Start Latency	First-time initialization of `all-MiniLM-L6-v2` takes 15–20s on CPU hosting environments.	Model quantization to **ONNX Runtime INT8** with persistent memory-mapped vector stores, reducing cold starts to $< 1.2\text{s}$.
3. Evaluation Microservice Cloud Cost	Full knowledge base audits using GPT-4o as judge incur token costs on large multi-thousand corpora.	Knowledge distillation into a lightweight **7B Small Language Model (SLM)** fine-tuned on code quality heuristics for offline execution.
4. Relational Database Session Volatility	In-memory fallback tracking resets if MySQL is ungracefully disconnected.	Deployment of **Redis persistent caching clusters** with Append-Only File (AOF) journaling for distributed state isolation.

FIGURE 10



5.2 Scaling Strategy: Democratizing AI for the Global South

FIGURE 11



1. **Zero-Token Sovereign Edge Deployment**: Deployable on commodity workstations via local Ollama weights and pre-seeded vector caches, providing offline prompt literacy training with zero cloud expenses.
2. **Federated Knowledge Harvesting**: Universities across the UNU Global AI Network can synchronize validated, self-healed prompt-code pairs via lightweight JSON diffs.
3. **Open-Source Pedagogical Reference**: Positioned as the open-source blueprint for responsible prompt engineering and AI literacy education in developing regions.

6. PROJECT ARTIFACTS & EVIDENCE REPOSITORY

System Layer	Operational Module / File Path	Technical Responsibility & Evidence Scope
Core Backend	`backend/main.py`	FastAPI application lifecycle, CORS policy, middleware, and OpenAPI 3.1 documentation endpoints.
Prompt Registry & Compression	`backend/core/prompts.py`	Mode-specific prompt harnesses, Bloom's cognitive tiering, CacheAligner prefix stabilization, and AST CodeCompressor.
Prompt Bounds	`backend/api/ai_chat.py`	200–2000 character C-I-O-E protocol validator and 60s deliberate reflection cooldown limiter.
Adaptive Gateway	`backend/services/adaptive_router.py`	3-tier cascade routing (Student Key → 6-Key Pool → Local Ollama Qwen2.5 14B) with Output Shaper.
Retrieval Engine	`backend/services/ai_service.py`	Dense-Sparse Hybrid Search (`all-MiniLM-L6-v2` + `BM25Okapi`), RRF fusion ($k = 60$), and fast-path gate ($s \geq 0.88$).
Sustainability Engine	`backend/services/sustainability.py`	Real-time physical telemetry formulation (Wh, $\mathrm{kg\ CO_2e}$, mL water) with Indonesian Grid factor ($\mathrm{CIF} = 0.384$).
Gamification Engine	`backend/services/gamification.py`	Peer-relative dynamic token threshold calculation ($\mathrm{Threshold} = 1.10 \times \overline{\mathrm{Usage}}_{\mathrm{peers}}$).
Data Access Layer	`backend/core/db.py`	MySQL connection pooling, circuit breakers, user API key isolation, and in-memory fallback queues.
Governance Pipeline	`code_evaluator_service/evaluator/evaluator_pipeline.py`	5-stage closed-loop quality scoring, duplicate filtering, and automated JSON pre-deletion backups.
Static Code Analysis	`code_evaluator_service/evaluator/static_analysis.py`	Abstract Syntax Tree (AST) validation, language detection, Radon Cyclomatic Complexity, and Maintainability Index.
Anomaly Detection	`code_evaluator_service/evaluator/anomaly_detection.py`	Scikit-learn Isolation Forest 5D unsupervised outlier screening model ($\gamma = 0.10$).
LLM-as-a-Judge	`code_evaluator_service/evaluator/llm_judge.py`	Multi-criteria snippet evaluation harness (A, L, Q) with heuristic fallback scoring.
Empirical Benchmarks	`pengujian semantic similarity/EXECUTIVE_SUMMARY.md`	200-query Gold Standard ground truth validation report ($\mathrm{MRR} = 1.000, \mathrm{Precision@1} = 100\%$).
Governance Case Run	`code_evaluator_service/reports/evaluation_report_20260312T094424Z.json`	Live audit dataset of 678 production entries (95.43% valid retention, 30 prunings, 1 quarantine).

System Layer	Operational Module / File Path	Technical Responsibility & Evidence Scope
LMS Chat Client	`estrange/v2/v2/ssparc/chat.php`	Production PHP interface with live cooldown timer, dynamic quota counter, and 200-char validation.
Academic Integrity	`estrange/v2/v2/student_assessment_submit.php`	Assignment submission engine with automated AST/token plagiarism detection ($\geq 70\%$ threshold).
Defense Adjudication	`estrange/v2/v2/student_assessment_submit_suspicious.php`	Structured student defense response workflow for lecturer resolution.
Peer Review Module	`estrange/v2/v2/student_peer_review.php`	Anonymized peer code evaluation and code clarity points allocation interface.

Conclusion & Multilateral Commitment

S-SPARC demonstrates that transforming the educational paradigm requires moving beyond unstructured conversational chatbots toward **calibrated, specific smart prompting grounded in Bloom's Taxonomy and Headroom-style context compression**. By enforcing the **200-character C-I-O-E protocol**, **AST CodeCompressor context reduction**, **CacheAligner KV-cache optimization**, **0-token semantic caching**, **autonomous closed-loop governance**, and **physical sustainability telemetry**, S-SPARC establishes a mature (TRL 7), youth-led standard for **AI Literacy and Sustainable Higher Education** under the auspices of the **United Nations University Global AI Network**.